V.F.S. v2.0 (Open-Gate)
Lyapunov Core Proof Formulae

with full term-by-term derivation and Sophianic Folding compatibility

Core extraction from `lyapunov.html`

Abstract

This hybrid document collects the core formulae of the Lyapunov proof for V.F.S. v2.0 (Open-Gate), preserves the original proof structure, and adds Sophianic Folding as a conservative Lyapunov-compatible refinement. It contains no graphs. It records the base dynamics, normalized variables, Lyapunov functional, active-domain walls, radial corridor, bounded Open-Gate source, memory bounds, non-Zeno estimate, and asymptotic cleansing statements. In addition, the central dissipative estimate $\dot{\mathcal{L}}_{\mathrm{VFS}} \le C - C_1\mathcal{L}_{\mathrm{VFS}}$ — previously stated as a boxed result — is here *derived term by term*: the contraction rate C_1 is built from exactly two sign-definite blocks (resistance cleansing and will–action alignment), while the Sophia, radial and Open-Gate contributions are bounded and enter only the constant C. The auxiliary light-filter constant is proved positive in closed form, so that $C_1 > 0$ holds for *every* admissible parameter set under the standing walls, not merely for the numerical witness. The added folding layer is treated as a morphological tension-relaxation mode: it may enlarge the bounded constant but does not replace the two sign-definite contracting blocks that generate C_1.

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1 Notation and Core Variables

Symbol	Meaning
V	Voluntas: will coordinate.
F	Factum: action coordinate.
σ	Singular resistance.
λ	Sophia: wisdom coordinate.
Ω_P	Brim / dynamic capacity of Pleroma.
u	Synergic intensity from V and F.
Λ_c	Transformation threshold.
μ	Saturation margin relative to the death-boundary.
q	Normalized Sophia, $q = \lambda/\Omega_P$.
r	Normalized intensity, $r = u/\Omega_P$.
h_1, h_2	Normalized will and action.
h_3	Normalized threshold variable.

Core threshold and intensity:

$$u = \sqrt{VF + \varepsilon}, \qquad \Lambda_c = \frac{\gamma}{\delta}, \qquad \varepsilon \geq 0. \tag{1}$$

Transformation/collapse reading:

$$u > \Lambda_c \quad \text{transformation region}, \qquad u < \Lambda_c \quad \text{collapse region}. \tag{2}$$

2 Base V.F.S. Dynamics

Resistance equation:

$$\dot{\sigma} = (\gamma - \delta u)\tanh(\kappa\sigma) = -\delta(u - \Lambda_c)\tanh(\kappa\sigma). \tag{3}$$

Symmetric will–action dynamics:

$$\dot{V} = \mu(aF + c\lambda) - \rho V, \qquad \dot{F} = \mu(aV + c\lambda) - \rho F, \qquad \mu = 1 - \frac{u}{\Omega_P}. \tag{4}$$

Capacity / Brim equation:

$$\dot{\Omega_P} = \alpha\lambda. \tag{5}$$

Closed Microcosm limit:

$$\dot{\lambda} = -\dot{\sigma} = \delta(u - \Lambda_c)\tanh(\kappa\sigma), \qquad \dot{\sigma} + \dot{\lambda} = 0. \tag{6}$$

Open-Gate Sophia dynamics:

$$\dot{\lambda} = -\dot{\sigma} + I_{\text{gate}}(t) = \delta(u - \Lambda_c)\tanh(\kappa\sigma) + I_{\text{gate}}(t). \tag{7}$$

Open balance:

$$\dot{\sigma} + \dot{\lambda} = I_{\mathrm{gate}}(t) \geq 0, \qquad \sigma(t) + \lambda(t) = \sigma_0 + \lambda_0 + \mathcal{G}_{\mathrm{recepta}}(t). \tag{8}$$

Received grace integral:

$$\mathcal{G}_{\mathrm{recepta}}(t) = \int_0^t I_{\mathrm{gate}}(s)\, ds, \qquad \dot{\mathcal{G}}_{\mathrm{recepta}} = I_{\mathrm{gate}}. \tag{9}$$

3 Open-Gate Balance with Embodied Sophia

The folding-refined core separates Sophia into available and embodied parts:

$$\lambda_{\mathrm{tot}} = \lambda + \lambda_{\mathrm{form}}. \tag{10}$$

The available part obeys

$$\dot{\lambda} = \dot{\lambda}^{(0)} - \eta_{\mathrm{fold}} Q_{\mathrm{fold}} \lambda, \qquad \dot{\lambda}^{(0)} = -\dot{\sigma} + I_{\mathrm{gate}}(t), \tag{11}$$

while the embodied component receives exactly the transferred portion:

$$\dot{\lambda}_{\mathrm{form}} = \eta_{\mathrm{fold}} Q_{\mathrm{fold}} \lambda. \tag{12}$$

Consequently the Open-Gate balance is preserved in refined form:

$$\boxed{\frac{d}{dt}(\sigma + \lambda + \lambda_{\mathrm{form}}) = I_{\mathrm{gate}}(t), \qquad \sigma(t) + \lambda(t) + \lambda_{\mathrm{form}}(t) = C_0 + \mathcal{G}_{\mathrm{recepta}}(t).} \tag{13}$$

Thus the negative feedback term in $\dot{\lambda}$ is not destruction of Sophia; it is embodiment into stable Vessel-form.

On a quasi-stationary folded branch,

$$Q_{\mathrm{fold}} = \frac{A_{\mathrm{fold}}}{B_{\mathrm{fold}}}, \tag{14}$$

and the available-Sophia damping is effectively increased:

$$\gamma_{\mathrm{eff}} = \gamma_\lambda + \eta_{\mathrm{fold}} Q_{\mathrm{fold}}. \tag{15}$$

Therefore

$$Q_{\mathrm{fold}} > 0 \quad \Longrightarrow \quad \dot{\lambda} < \dot{\lambda}^{(0)}. \tag{16}$$

This is the core feedback meaning of folding: it stabilizes Open-Gate excess by form-transformation.

4 Open-Gate Source

Positive part of Sophia:

$$\lambda_+ = \frac{1}{m} \ln(1 + e^{m\lambda}), \qquad m > 0. \tag{17}$$

Effective gate:

$$g_{\mathrm{eff}}(t) = e^{-\tau t} + \left(1 - e^{-\tau t}\right) \frac{H_K}{H_s + H_K}. \tag{18}$$

Live Open-Gate source:

$$I_{\text{gate}}(t) = \zeta_0\, g_{\text{eff}}(t) e^{-\phi\sigma(t)} \frac{1}{1+\chi\lambda_+(t)}. \tag{19}$$

Boundedness of the gate:

$$H_K \geq 0, \quad H_s > 0 \quad \implies \quad 0 \leq \frac{H_K}{H_s + H_K} < 1, \qquad 0 \leq g_{\text{eff}}(t) \leq 1. \tag{20}$$

Boundedness of live inflow:

$$0 \leq I_{\text{gate}}(t) \leq \zeta_0. \tag{21}$$

5 Resistance Potential

Resistance potential:

$$\Psi_\sigma(\sigma) = \int_0^\sigma \tanh(\kappa s)\, ds = \frac{1}{\kappa} \ln\cosh(\kappa\sigma) \geq 0. \tag{22}$$

Derivative of the resistance potential:

$$\dot{\Psi}_\sigma = \Psi'_\sigma(\sigma)\dot{\sigma} = \tanh(\kappa\sigma)(\gamma - \delta u)\tanh(\kappa\sigma) = (\gamma - \delta u)\tanh^2(\kappa\sigma). \tag{23}$$

Active transformation margin:

$$u \geq \Lambda_c + \eta, \qquad \eta > 0. \tag{24}$$

Dissipation in the active margin:

$$\dot{\Psi}_\sigma \leq -\delta\eta \tanh^2(\kappa\sigma). \tag{25}$$

On $0 \leq \sigma \leq C_\sigma$, there exists $c_\sigma > 0$ such that:

$$\dot{\Psi}_\sigma \leq -c_\sigma \Psi_\sigma. \tag{26}$$

Hence:

$$\Psi_\sigma(t) \leq \Psi_\sigma(0) e^{-c_\sigma t}. \tag{27}$$

Near $\sigma = 0$:

$$\Psi_\sigma(\sigma) \sim \frac{\kappa}{2}\sigma^2, \qquad \sigma(t) = O\big(e^{-c_\sigma t/2}\big). \tag{28}$$

6 Absolute Barrier Functional

The absolute barrier is a motivational, non-primary barrier guarding absolute boundary faces:

$$\begin{aligned}\mathcal{L}_{abs} = A_\sigma\Psi_\sigma + E_V V^2 + E_F F^2 + E_\lambda \lambda^2 + E_\Delta \frac{(V-F)^2}{2} \\ - \eta_V \ln V - \eta_F \ln F - \eta_\Omega \ln(\Omega_P - \Lambda_c) - \eta_m \ln(\Omega_P - u).\end{aligned} \tag{29}$$

Guarded absolute faces:

$$V = 0, \qquad F = 0, \qquad \Omega_P = \Lambda_c, \qquad u = \Omega_P. \tag{30}$$

7 Normalized Variables

Normalized coordinates:

$$h_1 = \frac{V}{\Omega_P}, \qquad h_2 = \frac{F}{\Omega_P}, \qquad q = \frac{\lambda}{\Omega_P}, \qquad r = \frac{u}{\Omega_P}, \qquad \mu = 1 - r, \qquad h_3 = 1 - \frac{\Lambda_c}{\Omega_P}. \tag{31}$$

Product relation:

$$r^2 = h_1 h_2 + e(h_3), \qquad e(h_3) = \varepsilon \frac{(1-h_3)^2}{\Lambda_c^2}. \tag{32}$$

Closed normalized dynamics:

$$\dot{h}_1 = \mu(a h_2 + c q) - \rho h_1 - \alpha q h_1, \tag{33}$$

$$\dot{h}_2 = \mu(a h_1 + c q) - \rho h_2 - \alpha q h_2, \tag{34}$$

$$\dot{q} = \delta(r + h_3 - 1)\tanh(\kappa\sigma) - \alpha q^2, \tag{35}$$

$$\dot{h}_3 = \alpha q(1 - h_3). \tag{36}$$

Open-Gate normalized q-source:

$$S_q = \frac{I_{\text{gate}}}{\Omega_P} = \frac{\zeta_0 g_{\text{eff}} e^{-\phi\sigma}}{(1+\chi\lambda_+)\Omega_P}. \tag{37}$$

Open-Gate normalized q-equation:

$$\dot{q} = \delta(r + h_3 - 1)\tanh(\kappa\sigma) - \alpha q^2 + S_q. \tag{38}$$

Bound on S_q inside the active domain:

$$0 \le S_q \le \frac{\zeta_0}{\Omega_P} \le \frac{\zeta_0(1 - h_{3,*})}{\Lambda_c} =: S_q^{\max}. \tag{39}$$

8 Product-Balance Mechanism

Balance defect:

$$D_\Delta = \frac{1}{2}(h_1 - h_2)^2. \tag{40}$$

Let:

$$d = h_1 - h_2. \tag{41}$$

Difference equation:

$$\dot{d} = -(a\mu + \rho + \alpha q)d. \tag{42}$$

Dissipation of balance defect:

$$\dot{D}_\Delta = -2(a\mu + \rho + \alpha q)D_\Delta. \tag{43}$$

If $\mu \ge \mu_* > 0$ and $a > 0$:

$$\dot{D}_\Delta \le -2a\mu_* D_\Delta. \tag{44}$$

Exponential will–action alignment:

$$D_\Delta(t) \le D_\Delta(0)e^{-2a\mu_* t}, \qquad |h_1(t) - h_2(t)| \le |h_1(0) - h_2(0)|e^{-a\mu_* t}. \tag{45}$$

Product floor from radial and threshold walls:

$$\Omega_P \ge \Omega_{\min} := \frac{\Lambda_c}{1 - h_{3,*}}. \tag{46}$$

If $r \ge r_*$ and $h_3 \ge h_{3,*}$:

$$h_1 h_2 = r^2 - \frac{\varepsilon}{\Omega_P^2} \ge p_* := r_*^2 - \frac{\varepsilon}{\Omega_{\min}^2}. \tag{47}$$

Let:

$$d_* := \sqrt{2D^*}. \tag{48}$$

Product-balance lower bound:

$$h_1, h_2 \ge m_{pb} := \frac{\sqrt{d_*^2 + 4p_*} - d_*}{2} > 0. \tag{49}$$

9 Lyapunov Functional

Product-balance Lyapunov core:

$$\mathcal{L}_{pb} = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2. \tag{50}$$

Radial logarithmic barrier:

$$B(r) = -\ln(1 - r), \qquad 0 < r < 1, \qquad r = \frac{u}{\Omega_P}. \tag{51}$$

Non-normalized radial barrier:

$$B(u, \Omega_P) = -\ln\left(1 - \frac{u}{\Omega_P}\right) = \ln\left(\frac{\Omega_P}{\Omega_P - u}\right). \tag{52}$$

Primary V.F.S. Lyapunov functional:

$$\boxed{\mathcal{L}_{\mathrm{VFS}} = \mathcal{L}_{pb} + A_r B(r) = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2 + A_r B(r), \qquad A_\sigma, A_\Delta, A_q, A_r > 0.} \tag{53}$$

Expanded form:

$$\mathcal{L}_{\mathrm{VFS}} = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2 - A_r \ln(1 - r). \tag{54}$$

Non-normalized expanded form:

$$\mathcal{L}_{\mathrm{VFS}} = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2 - A_r \ln\left(1 - \frac{u}{\Omega_P}\right). \tag{55}$$

Radial boundary protection:

$$B(r) \to +\infty \quad \text{as} \quad r \to 1^-. \tag{56}$$

10 Sophianic Folding Extension of the Lyapunov Functional

10.1 Status of the extension

The preceding functional $\mathcal{L}_{\mathrm{VFS}}$ remains the primary Open-Gate Lyapunov certificate. Sophianic Folding is added as a conservative refinement: it does not replace the resistance-cleansing and will–action alignment mechanisms, but adds a morphological relaxation channel for excess Sophia. Throughout this section, $q = \lambda/\Omega_P$ remains the normalized Sophia variable used above, while q_{fold} denotes a distinct shape-mode amplitude.

The new variables are

$$q_{\mathrm{fold}}(t) \quad \text{(folding amplitude)}, \qquad Q_{\mathrm{fold}}(t) := q_{\mathrm{fold}}(t)^2 \quad \text{(folding intensity)}, \tag{57}$$

$$\lambda_{\mathrm{form}}(t) \quad \text{(Sophia embodied into Vessel-form)}. \tag{58}$$

The unrefined Open-Gate Lyapunov model is recovered by setting

$$q_{\mathrm{fold}} = 0, \qquad Q_{\mathrm{fold}} = 0, \qquad \lambda_{\mathrm{form}} = 0. \tag{59}$$

10.2 Folding activation and normal form

Let the pre-folding Sophia production be

$$\dot{\lambda}^{(0)} = -\dot{\sigma} + I_{\mathrm{gate}}(t). \tag{60}$$

The folding activation index is

$$A_{\mathrm{fold}} = c_0\lambda + c_1\dot{\lambda}^{(0)} - a_0, \qquad a_0, c_0, c_1 > 0, \tag{61}$$

and the stabilizing quartic coefficient is

$$B_{\mathrm{fold}} = b + 2c_1\eta_{\mathrm{fold}}\lambda, \qquad b > 0, \qquad \eta_{\mathrm{fold}} > 0. \tag{62}$$

Hence $B_{\mathrm{fold}} > 0$ whenever $\lambda \geq 0$. The folding dynamics is the one-mode gradient normal form

$$\dot{q}_{\mathrm{fold}} = A_{\mathrm{fold}}q_{\mathrm{fold}} - B_{\mathrm{fold}}q_{\mathrm{fold}}^3. \tag{63}$$

If $A_{\mathrm{fold}} < 0$, the old Vessel-form $q_{\mathrm{fold}} = 0$ is stable. If $A_{\mathrm{fold}} > 0$, the stable folded branches satisfy

$$q_{\mathrm{fold},\pm}^2 = Q_{\mathrm{fold}}^* = \frac{A_{\mathrm{fold}}}{B_{\mathrm{fold}}}. \tag{64}$$

10.3 Morphological tension potential

The folding potential is not physical gravitational energy. It is a Lyapunov/shape potential measuring morphological tension of the Vessel-form:

$$\mathcal{E}_{\mathrm{fold}}(q_{\mathrm{fold}}) = -\frac{1}{2}A_{\mathrm{fold}}q_{\mathrm{fold}}^2 + \frac{1}{4}B_{\mathrm{fold}}q_{\mathrm{fold}}^4. \tag{65}$$

Then

$$\partial_{q_{\mathrm{fold}}}\mathcal{E}_{\mathrm{fold}} = -A_{\mathrm{fold}}q_{\mathrm{fold}} + B_{\mathrm{fold}}q_{\mathrm{fold}}^3, \tag{66}$$

and therefore

$$\dot{q}_{\mathrm{fold}} = -\partial_{q_{\mathrm{fold}}}\mathcal{E}_{\mathrm{fold}}. \tag{67}$$

For frozen or adiabatically varying Open-Gate parameters,

$$\boxed{\dot{\mathcal{E}}_{\text{fold}} = -\left(\partial_{q_{\text{fold}}}\mathcal{E}_{\text{fold}}\right)^2 = -\dot{q}_{\text{fold}}^2 \le 0.} \tag{68}$$

Thus folding is an energy-descending, tension-relaxing mode rather than a singular loss of stability.

10.4 Shifted non-negative folding functional

For Lyapunov use, the negative minimum of $\mathcal{E}_{\text{fold}}$ is shifted to zero. When $A_{\text{fold}} > 0$,

$$\mathcal{E}_{\text{fold,min}} = -\frac{A_{\text{fold}}^2}{4B_{\text{fold}}}, \tag{69}$$

so that

$$\mathcal{V}_{\text{fold}} := \mathcal{E}_{\text{fold}} - \mathcal{E}_{\text{fold,min}} = \frac{B_{\text{fold}}}{4}\left(q_{\text{fold}}^2 - \frac{A_{\text{fold}}}{B_{\text{fold}}}\right)^2 \ge 0. \tag{70}$$

When $A_{\text{fold}} \le 0$, one may simply take

$$\mathcal{V}_{\text{fold}} = \frac{1}{2}|A_{\text{fold}}|q_{\text{fold}}^2 + \frac{1}{4}B_{\text{fold}}q_{\text{fold}}^4 \ge 0, \tag{71}$$

with the minimum at $q_{\text{fold}} = 0$.

10.5 Extended Lyapunov candidate

The folding-compatible Lyapunov candidate is

$$\boxed{\mathcal{L}_{\text{ext}} = \mathcal{L}_{\text{VFS}} + w_{\text{fold}}\mathcal{V}_{\text{fold}}, \qquad w_{\text{fold}} > 0.} \tag{72}$$

This construction preserves the original $\mathcal{L}_{\text{VFS}}$ as the core certificate, while adding a non-negative measure of distance from the stable folded morphology.

11 Active Domain

Active domain:

$$\mathcal{D}_A = \left\{\begin{array}{l} 0 \le \sigma \le C_\sigma, \quad q \in [0, q^*], \quad h_3 \in [h_{3,*}, 1), \\ r \in [r_*, r^*], \quad \mu = 1 - r \ge \mu_* > 0, \\ D_\Delta \le D^*, \quad h_1 > 0, \quad h_2 > 0 \end{array}\right\}, \qquad 0 < r_* < r^* < 1. \tag{73}$$

Equivalent death-boundary avoidance inside $\mathcal{D}_A$:

$$0 < r < 1 \qquad \Longleftrightarrow \qquad 0 < u < \Omega_P. \tag{74}$$

Closed upper q-wall:

$$\alpha(q^*)^2 \ge \delta r^*. \tag{75}$$

Open-Gate upper q-wall:

$$\boxed{\alpha(q^*)^2 \ge \delta r^* + S_q^{\max}.} \tag{76}$$

Lower q-wall in Open-Gate:

$$q = 0: \qquad \dot{q} = \delta(r + h_3 - 1)\tanh(\kappa\sigma) + S_q \ge 0 \tag{77}$$

provided the active margin makes $r + h_3 - 1 \ge 0$.

12 Radial Corridor

Radial identity from $r^2 = h_1 h_2 + e(h_3)$:

$$2r\dot{r} = \mu\big[a(h_1^2 + h_2^2) + cq(h_1 + h_2)\big] - 2\rho(r^2 - e(h_3)) - 2\alpha q r^2 =: \mathcal{R}_{\mathrm{rad}}. \tag{78}$$

Maximum e on the active corridor:

$$e_*^{\max} = \varepsilon \frac{(1 - h_{3,*})^2}{\Lambda_c^2}. \tag{79}$$

Non-degeneracy:

$$r_*^2 > e_*^{\max}. \tag{80}$$

Active margin:

$$r_* + h_{3,*} - 1 \geq \eta \frac{1 - h_{3,*}}{\Lambda_c}. \tag{81}$$

Lower radial wall:

$$a(1 - r_*)(r_*^2 - e_*^{\max}) \geq (\rho + \alpha q^*) r_*^2. \tag{82}$$

This wall is sufficient, not tight. The exact lower-wall requirement is

$$(1 - r_*) a (r_*^2 - e) \geq \rho(r_*^2 - e) + \alpha q r_*^2, \tag{83}$$

and the stated form replaces $\rho(r_*^2 - e)$ by the larger ρr_*^2.

Upper radial wall:

$$(1 - r^*)\big[a(d_*^2 + 2r^{*2}) + cq^* \sqrt{d_*^2 + 4r^{*2}}\big] \leq 2\rho(r^{*2} - e_*^{\max}). \tag{84}$$

Direct upper radial inwardness condition:

$$\mu\left[a(h_1^2 + h_2^2) + cq(h_1 + h_2)\right] \leq 2\rho(r^{*2} - e) + 2\alpha q r^{*2} \qquad \text{on } r = r^*. \tag{85}$$

Open-Gate radial convention:

$$q_{\mathrm{new}}^* := \sqrt{\frac{\delta r^* + S_q^{\max}}{\alpha}}. \tag{86}$$

Conservative lower radial wall with q_{new}^*:

$$a(1 - r_*)(r_*^2 - e_*^{\max}) \geq (\rho + \alpha q_{\mathrm{new}}^*) r_*^2. \tag{87}$$

Epektasis-dominance derivative at the upper radial wall:

$$\left.\frac{\partial \mathcal{R}_{\mathrm{rad}}}{\partial q}\right|_{r^*} = \mu c(h_1 + h_2) - 2\alpha r^{*2}. \tag{88}$$

Worst-case bound:

$$\left.\frac{\partial \mathcal{R}_{\mathrm{rad}}}{\partial q}\right|_{r^*} \leq (1 - r^*) c \sqrt{d_*^2 + 4r^{*2}} - 2\alpha r^{*2}. \tag{89}$$

Epektasis-dominance sufficient condition:

$$\boxed{2\alpha r^{*2} \geq (1 - r^*) c \sqrt{d_*^2 + 4r^{*2}} \quad \Longrightarrow \quad \frac{\partial \mathcal{R}_{\mathrm{rad}}}{\partial q} \leq 0.} \tag{90}$$

13 Lyapunov Dissipative Estimate

This section derives the estimate $\dot{\mathcal{L}}_{\mathrm{VFS}} \le C - C_1\mathcal{L}_{\mathrm{VFS}}$ term by term and exhibits an explicit, strictly positive C_1.

13.1 Standing admissibility data

We use the conditions that already define and protect the active domain $\mathcal{D}_A$:

$$\text{(A1) } a>0,\ \delta>0,\ \gamma>0; \qquad \text{(A2) } 0\le h_{3,*}<1,\ 0<r_*<r^*<1,\ \mu_*:=1-r^*>0; \tag{91}$$

$$\text{(A3) } m_*:=r_*+h_{3,*}-1>0 \ \text{ (active margin)}; \qquad \text{(A4) } \kappa>0,\ 0<C_\sigma<\infty. \tag{92}$$

Inside $\mathcal{D}_A$: $\sigma\in[0,C_\sigma]$, $q\in[0,q^*]$, $h_3\in[h_{3,*},1)$, $r\in[r_*,r^*]$, $D_\Delta\le D^*$, and $0\le S_q\le S_q^{\max}$. Write

$$T:=\tanh(\kappa\sigma)\in[0,1], \qquad \Omega_P=\frac{\Lambda_c}{1-h_3}, \qquad m:=r+h_3-1=\frac{u-\Lambda_c}{\Omega_P}\ge m_*>0, \tag{93}$$

so that the resistance equation reads $\dot\sigma=(\gamma-\delta u)T=-\delta\Omega_P m T$.

13.2 Derivative decomposition

With $\mathcal{L}_{\mathrm{VFS}} = A_\sigma\Psi_\sigma(\sigma)+A_\Delta D_\Delta+A_q q^2+A_r B(r)$, $B(r)=-\ln(1-r)$, and $\Psi_\sigma'(\sigma)=T$, $B'(r)=1/(1-r)$,

$$\dot{\mathcal{L}}_{\mathrm{VFS}} = \underbrace{A_\sigma T\dot\sigma}_{\text{(I)}}+\underbrace{A_\Delta\dot D_\Delta}_{\text{(II)}}+\underbrace{2A_q q\dot q}_{\text{(III)}}+\underbrace{A_r\frac{\dot r}{1-r}}_{\text{(IV)}}. \tag{94}$$

13.3 Block (I): resistance — contracting

Substituting $\dot\sigma=-\delta\Omega_P m T$,

$$\text{(I)} = -A_\sigma\,\delta\,\Omega_P\,m\,T^2\le 0. \tag{95}$$

By Lemma 1 below $T^2\ge c_\Psi\Psi_\sigma$ on $[0,C_\sigma]$, and $\Omega_P\ge\Omega_{\min}:=\Lambda_c/(1-h_{3,*})$, $m\ge m_*$. Hence

$$\boxed{\text{(I)}\le-\kappa_\sigma A_\sigma\Psi_\sigma, \qquad \kappa_\sigma:=\delta\,\Omega_{\min}\,m_*\,c_\Psi.} \tag{96}$$

13.4 Block (II): balance — contracting

From $\dot D_\Delta=-2(a\mu+\rho+\alpha q)D_\Delta$ and $a\mu+\rho+\alpha q\ge a\mu_*$,

$$\boxed{\text{(II)}\le-\kappa_\Delta A_\Delta D_\Delta, \qquad \kappa_\Delta:=2a\mu_*.} \tag{97}$$

13.5 Block (III): Sophia — bounded

Expanding the normalized q-dynamics $\dot q=\delta mT-\alpha q^2+S_q$,

$$\frac{d}{dt}(A_q q^2)=2A_q q\dot q=-2A_q\alpha q^3+\underbrace{2A_q\,\delta mT\,q}_{\text{transfer}}+\underbrace{2A_q S_q\,q}_{\text{gate}}. \tag{98}$$

The transfer term shares the factor δmT with block (I): the conversion that *drains σ feeds q.* Both positive terms are bounded on $\mathcal{D}_A$,

$$0 \le 2A_q\delta mTq \le 2A_q\delta m^{\max}q^*, \qquad 0 \le 2A_qS_qq \le 2A_qq^*S_q^{\max} =: \Delta C_{\text{gate}}, \tag{99}$$

while $-2A_q\alpha q^3 \le 0$. Since $C_1q^2 - 2A_q\alpha q^3$ is bounded above on the compact $[0, q^*]$ for any C_1, block (III) obeys

$$\text{(III)} \le -C_1A_qq^2 + B_q, \qquad B_q < \infty; \tag{100}$$

it contributes only to the constant.

13.6 Block (IV): radial — bounded

$B(r) = -\ln(1-r)$ diverges only as $r \to 1^-$, but $\mathcal{D}_A$ caps $r \le r^* < 1$, so $B(r) \le -\ln(1-r^*) < \infty$. With $r \ge r_* > 0$, $\mu \ge \mu_* > 0$ and bounded h_1, h_2, the radial rate $\dot{r} = \mathcal{R}_{\text{rad}}/(2r)$ is bounded, so block (IV) is bounded on $\mathcal{D}_A$ and contributes only to the constant. Its sign on the face $r = r^*$ (i.e. $\mathcal{R}_{\text{rad}} \le 0$, the upper radial wall) is used for invariance, not for C_1.

13.7 The light-filter constant

Lemma 1. *For every $\kappa > 0$ and $C_\sigma < \infty$,*

$$c_\Psi := \min_{0\le\sigma\le C_\sigma} \frac{\tanh^2(\kappa\sigma)}{\Psi_\sigma(\sigma)} = \kappa\,\frac{\tanh^2(\kappa C_\sigma)}{\ln\cosh(\kappa C_\sigma)} > 0, \qquad \Psi_\sigma(\sigma) = \tfrac{1}{\kappa}\ln\cosh(\kappa\sigma). \tag{101}$$

Proof. With $t = \kappa\sigma$ the ratio equals $\kappa\, g(t)$, $g(t) = \tanh^2 t/\ln\cosh t$ on $(0, \kappa C_\sigma]$. As $t \to 0^+$, $\tanh^2 t \sim t^2$ and $\ln\cosh t \sim t^2/2$, so $g(t) \to 2$ and g extends continuously to $g(0) = 2$; moreover $g > 0$ on $(0, \infty)$ since $\tanh^2 t > 0$, $\ln\cosh t > 0$. The sign of g' equals that of $h(t) := 2\,\mathrm{sech}^2 t\, \ln\cosh t - \tanh^2 t$. Now $h(0) = 0$ and

$$h'(t) = -4\,\mathrm{sech}^2 t\, \tanh t\, \ln\cosh t < 0 \quad (t > 0), \tag{102}$$

so $h(t) < 0$ for $t > 0$, whence $g' < 0$ and g is strictly decreasing on $(0, \infty)$. The minimum over $[0, \kappa C_\sigma]$ is thus attained at the right endpoint, giving the closed form, which is positive. □

13.8 Dissipative estimate

Proposition 1 (Dissipative estimate with explicit C_1)**.** *Under* (A1)–(A4) *there exist $C_1 > 0$ and $C < \infty$ with*

$$\boxed{\dot{\mathcal{L}}_{\text{VFS}} \le C - C_1\mathcal{L}_{\text{VFS}} \quad \textit{on } \mathcal{D}_A, \qquad C_1 = \min\{\kappa_\sigma, \kappa_\Delta\} = \min\{\delta\,\Omega_{\min}\, m_*\, c_\Psi,\ 2a\mu_*\} > 0.} \tag{103}$$

The Open-Gate source affects only C through $0 \le \Delta C_{\text{gate}} \le 2A_qq^*S_q^{\max}$ and leaves C_1 unchanged.

Proof. Summing (I)–(IV),

$$\dot{\mathcal{L}}_{\text{VFS}} \le -\kappa_\sigma A_\sigma\Psi_\sigma - \kappa_\Delta A_\Delta D_\Delta + C', \tag{104}$$

with $C' < \infty$ collecting the bounded blocks (III)–(IV) (transfer, gate, radial). Set $C_1 = \min\{\kappa_\sigma, \kappa_\Delta\}$, so that $-\kappa_\sigma A_\sigma\Psi_\sigma - \kappa_\Delta A_\Delta D_\Delta \le -C_1(A_\sigma\Psi_\sigma + A_\Delta D_\Delta)$. Since $A_qq^2 + A_rB(r) \le P_{\max} < \infty$ on $\mathcal{D}_A$,

$$\dot{\mathcal{L}}_{\text{VFS}} \le -C_1\mathcal{L}_{\text{VFS}} + C_1P_{\max} + C' =: -C_1\mathcal{L}_{\text{VFS}} + C. \tag{105}$$

Finiteness of C. Blocks (III),(IV) are bounded; the only block unbounded as $h_3 \to 1^-$ ($\Omega_P \to \infty$) is (I)$= -A_\sigma \delta \Omega_P m T^2 \leq 0$; hence $\sup_{\mathcal{D}_A}(\dot{\mathcal{L}_{\mathrm{VFS}}} + C_1 \mathcal{L}_{\mathrm{VFS}}) < \infty$. *Positivity of C_1.* $\kappa_\Delta = 2a\mu_* > 0$ by (A1),(A2); and $\kappa_\sigma = \delta\, \Omega_{\min}\, m_*\, c_\Psi > 0$ because $\delta > 0$ (A1), $\Omega_{\min} = \Lambda_c/(1-h_{3,*}) > 0$ (A2), $m_* > 0$ (A3) and $c_\Psi > 0$ (Lemma 1 with (A4)). Each factor is one of the standing admissibility data, so $C_1 > 0$ for every admissible parameter set. □

Open-Gate constant split:

$$C = C_{\mathrm{closed}} + \Delta C_{\mathrm{gate}}, \qquad 0 \leq \Delta C_{\mathrm{gate}} \leq 2A_q q^* S_q^{\max}. \tag{106}$$

Key structural point. The dissipation rate C_1 is produced solely by the two sign-definite contracting modes — resistance cleansing (I) at rate κ_σ and will–action alignment (II) at rate κ_Δ. Sophia/Epektasis (q^2), radial position ($B(r)$) and received grace (S_q) cannot accelerate the decay of $\mathcal{L}_{\mathrm{VFS}}$; they only enlarge the bounded constant C. Thus C_1 is unchanged by the Open-Gate source, while C receives only the bounded contribution ΔC_{gate}.

Remark (No uniform floor at the admissibility boundary)*.* Positivity is per parameter set, not uniform: $\inf C_1 = 0$ over the whole admissible set, approached as $m_* \to 0^+$ (a set on the active-margin wall) or $\mu_* \to 0^+$ ($r^* \to 1$). A configuration pinned on the active margin has zero cleansing rate. The correct statement is "$C_1 > 0$ for each admissible set," not a single $C_1 > 0$ valid for all sets at once.

Remark (Relation to the numerical witness)*.* At the witness point of Section 20 (with $\kappa = 1$, $C_\sigma = 2$, unit weights) one gets $c_\Psi = 0.701$, $\kappa_\sigma = 0.255$, $\kappa_\Delta = 0.755$, hence $C_1 = 0.255$ (resistance-limited), versus an empirical envelope slope ≈ 1.5; the analytic rate is the conservative guaranteed value, as expected.

14 Folding-Compatible Dissipative Estimate

This section explains why the addition of Sophianic Folding does not destroy the Open-Gate Lyapunov estimate. The old estimate

$$\dot{\mathcal{L}_{\mathrm{VFS}}} \leq C - C_1 \mathcal{L}_{\mathrm{VFS}} \tag{107}$$

remains the base certificate. Folding contributes an additional non-negative shape functional whose own flow is descending.

Proposition 2 (Lyapunov compatibility of Sophianic Folding)**.** *Assume that the trajectory remains in the regular folding domain*

$$0 \leq Q_{\mathrm{fold}} \leq Q_{\max} < \infty, \qquad B_{\mathrm{fold}} \geq B_{\min} > 0, \tag{108}$$

and that A_{fold} and B_{fold} are frozen or vary adiabatically on the folding relaxation scale. Then the extended functional

$$\mathcal{L}_{\mathrm{ext}} = \mathcal{L}_{\mathrm{VFS}} + w_{\mathrm{fold}} \mathcal{V}_{\mathrm{fold}} \tag{109}$$

satisfies a dissipative bound of the same form

$$\boxed{\dot{\mathcal{L}}_{\mathrm{ext}} \leq C_{\mathrm{ext}} - C_1 \mathcal{L}_{\mathrm{ext}}} \tag{110}$$

on the regular folding domain, for some finite constant C_{ext}.

Proof. For frozen folding parameters,

$$\dot{\mathcal{V}}_{\text{fold}} = \dot{\mathcal{E}}_{\text{fold}} = -\left(\partial_{q_{\text{fold}}}\mathcal{E}_{\text{fold}}\right)^2 \leq 0. \tag{111}$$

Hence

$$\dot{\mathcal{L}}_{\text{ext}} = \dot{\mathcal{L}_{\text{VFS}}} + w_{\text{fold}}\dot{\mathcal{V}}_{\text{fold}} \leq \dot{\mathcal{L}_{\text{VFS}}} \leq C - C_1\mathcal{L}_{\text{VFS}}. \tag{112}$$

On the regular folding domain, $\mathcal{V}_{\text{fold}}$ is bounded above by a finite constant M_{fold}. Therefore

$$\mathcal{L}_{\text{VFS}} = \mathcal{L}_{\text{ext}} - w_{\text{fold}}\mathcal{V}_{\text{fold}} \geq \mathcal{L}_{\text{ext}} - w_{\text{fold}}M_{\text{fold}}. \tag{113}$$

Substitution gives

$$\dot{\mathcal{L}}_{\text{ext}} \leq C - C_1\mathcal{L}_{\text{ext}} + C_1 w_{\text{fold}}M_{\text{fold}} = C_{\text{ext}} - C_1\mathcal{L}_{\text{ext}}, \tag{114}$$

where $C_{\text{ext}} := C + C_1 w_{\text{fold}}M_{\text{fold}} < \infty$. For slowly moving parameters, extra terms

$$-\frac{1}{2}q_{\text{fold}}^2\dot{A}_{\text{fold}} + \frac{1}{4}q_{\text{fold}}^4\dot{B}_{\text{fold}} \tag{115}$$

are bounded on the regular domain and are absorbed into C_{ext}. Thus the contracting rate C_1 is not replaced by folding; only the bounded constant is enlarged. □

Interpretation. Sophianic Folding is a Lyapunov-compatible morphological stabilization. It does not generate a new death-boundary, and it is not a Perelman-type singular surgery. It is a regular form-transition in which excess available Sophia is relaxed into a lower-tension Vessel morphology.

15 Main Theorem Formulae

Positive invariance:

$$x(0) \in \mathcal{D}_A \quad \Longrightarrow \quad x(t) \in \mathcal{D}_A \quad \forall t \geq 0. \tag{116}$$

Reset admissibility holds by the Reset Lemma, with

$$L_R = \sup_{\mathcal{D}_A} \mathcal{L}_{\text{VFS}}. \tag{117}$$

Bound from the Lyapunov inequality:

$$\mathcal{L}_{\text{VFS}}(t) \leq L_* := \max\left\{L_R, \frac{C}{C_1}\right\}. \tag{118}$$

Radial barrier implication:

$$A_r B(r) \leq L_*. \tag{119}$$

Explicit radial bound:

$$-\ln(1-r) \leq \frac{L_*}{A_r} \quad \Longrightarrow \quad r(t) \leq 1 - e^{-L_*/A_r} < 1. \tag{120}$$

Death-boundary avoidance:

$$r(t) < 1 \quad \Longrightarrow \quad u(t) < \Omega_P(t). \tag{121}$$

Product-balance positivity:

$$h_1, h_2 \geq m_{pb} > 0. \tag{122}$$

No artificial q-floor:

$$q \geq 0 \quad \text{suffices; no assumption } q \geq q_0 > 0 \text{ is required.} \tag{123}$$

Folding-compatible extension:

$$q_{\text{fold}} = 0 \quad \text{recovers the original theorem,} \qquad q_{\text{fold}} \neq 0 \quad \text{adds a bounded shape-relaxation channel.} \tag{124}$$

16 Reset Admissibility (Resurrectio Jump)

The resets $\mathfrak{R}$ are *exogenous*: since $\mathcal{D}_A$ is positively invariant, the continuous flow never reaches the boundary of the active domain by itself. A death-like strike is therefore modeled as occurring at an interior point

$$x^- = (\sigma^-, \lambda^-, V, F, \Omega_P^-) \in \mathcal{D}_A \tag{125}$$

at an unpredictable time.

Reset map (linear metabolism). With $0 \leq q_R < 1$ and $\kappa_R > 0$,

$$\mathfrak{R}: \quad \sigma^+ = q_R \sigma^-, \qquad \lambda^+ = \lambda^- + (1-q_R)\sigma^-, \qquad (V, F) \text{ continuous}, \qquad \Omega_P^+ = \Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-. \tag{126}$$

Here

$$\mathcal{G}_{\text{recepta}}^- := \mathcal{G}_{\text{recepta}}(t_{\text{death}}^-), \qquad u^+ = \sqrt{VF + \varepsilon} = u^-. \tag{127}$$

Quantities.

$$A_R := \mathcal{G}_{\text{recepta}}^-, \qquad R_c := C_\sigma - \sigma_0 - \lambda_0, \tag{128}$$

$$R_{\min}^{\text{ceil}} := \frac{(1-q_R)\sigma^-}{q^* \kappa_R}, \qquad R_{\max} := \frac{\Omega_P^-(r^- - r_*)}{\kappa_R r_*}. \tag{129}$$

By the Open-Gate balance

$$\sigma^- + \lambda^- = \sigma_0 + \lambda_0 + \mathcal{G}_{\text{recepta}}^-, \tag{130}$$

one has

$$\mathcal{G}_{\text{recepta}}^- \geq R_c \quad \Longleftrightarrow \quad \sigma^- + \lambda^- \geq C_\sigma \quad \Longleftrightarrow \quad \lambda^- \geq C_\sigma - \sigma^-. \tag{131}$$

Lemma (reset admissibility). Let $\kappa_R > 0$ and $x^- \in \mathcal{D}_A$. If

$$\max\{R_c, R_{\min}^{\text{ceil}}\} \leq \mathcal{G}_{\text{recepta}}^- \leq R_{\max}, \tag{132}$$

then $x^+ := \mathfrak{R}(x^-) \in \mathcal{D}_A$ and

$$\mathcal{L}_{\text{VFS}}(x^+) \leq L_R := \sup_{\mathcal{D}_A} \mathcal{L}_{\text{VFS}} < \infty. \tag{133}$$

Proof. Coordinate by coordinate,

$$\sigma^+ = q_R \sigma^- \in [0, C_\sigma). \tag{134}$$

Moreover,

$$h_3^+ = 1 - \frac{\Lambda_c}{\Omega_P^+} \geq h_3^- \geq h_{3,*}, \qquad h_3^+ < 1, \tag{135}$$

since $\Omega_P^+ \geq \Omega_P^-$. Also

$$h_1^+ = h_1^- \frac{\Omega_P^-}{\Omega_P^+}, \qquad h_2^+ = h_2^- \frac{\Omega_P^-}{\Omega_P^+}, \tag{136}$$

so

$$D_\Delta^+ = \frac{1}{2}\left(d^- \frac{\Omega_P^-}{\Omega_P^+}\right)^2 \leq D_\Delta^- \leq D^*, \qquad h_1^+, h_2^+ > 0. \tag{137}$$

The radial coordinate satisfies

$$r^+ = \frac{r^- \Omega_P^-}{\Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-} \leq r^- \leq r^*. \tag{138}$$

The lower radial wall is preserved exactly when

$$r^+ \geq r_* \iff \kappa_R \mathcal{G}_{\text{recepta}}^- \leq \frac{\Omega_P^-(r^- - r_*)}{r_*} \iff \mathcal{G}_{\text{recepta}}^- \leq R_{\max}. \tag{139}$$

Finally,

$$q^+ = \frac{\lambda^- + (1 - q_R)\sigma^-}{\Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-} \geq 0. \tag{140}$$

Since $x^- \in \mathcal{D}_A$ gives $\lambda^- \leq q^* \Omega_P^-$,

$$\begin{aligned} q^*(\Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-) - (\lambda^- + (1 - q_R)\sigma^-) &\geq q^* \kappa_R \mathcal{G}_{\text{recepta}}^- - (1 - q_R)\sigma^- \\ &\geq 0 \end{aligned} \tag{141}$$

whenever $\mathcal{G}_{\text{recepta}}^- \geq R_{\min}^{\text{ceil}}$. Hence $q^+ \leq q^*$. Thus $x^+ \in \mathcal{D}_A$. The functional $\mathcal{L}_{\text{VFS}}$ is bounded on $\mathcal{D}_A$ because

$$\Psi_\sigma \leq \Psi_\sigma(C_\sigma), \qquad D_\Delta \leq D^*, \qquad q^2 \leq (q^*)^2, \qquad B(r) \leq -\ln(1 - r^*) < \infty. \tag{142}$$

Therefore $\mathcal{L}_{\text{VFS}}(x^+) \leq L_R$. □

Consequences.

(1) The admissible window floor is

$$\max\{R_c, R_{\min}^{\text{ceil}}\}. \tag{143}$$

It combines the theological gate and the geometric ceiling-floor: grace must expand the vessel sufficiently to absorb the converted resistance. For cleansed deaths, $\sigma^- \to 0$, the ceiling-floor vanishes and R_c alone remains binding.

(2) The condition $\kappa_R > 0$ suffices. A natural design choice is

$$\kappa_R \geq \frac{1}{q^*}, \tag{144}$$

so that the asymptotic ceiling $q^+ \to 1/\kappa_R \leq q^*$ and, in the intended window, $R_{\min}^{\text{ceil}} \leq R_c$.

(3) The admissible grace window is

$$\mathcal{G}_{\text{recepta}}^- \in \left[\max\{R_c, R_{\min}^{\text{ceil}}\}, R_{\max}\right]. \tag{145}$$

Below this window one obtains final death; above it one obtains over-dilution, $r < r_*$. The window is nonempty if and only if

$$\Omega_P^-(r^- - r_*) \geq \kappa_R r_* \max\{R_c, R_{\min}^{\text{ceil}}\}. \tag{146}$$

A death pinned at $r^- = r_*$ is not resurrectible.

(4) With the non-Zeno dwell time

$$\Delta t_j \geq \frac{\bar{h}_*}{\bar{M}_*} > 0, \tag{147}$$

every life-arc satisfies

$$\mathcal{L}_{\mathrm{VFS}} \leq L_* := \max\left\{L_R, \frac{C}{C_1}\right\}, \tag{148}$$

while admissible reset states themselves satisfy $\mathcal{L}_{\mathrm{VFS}}(x^+) \leq L_R$. Resets do not accumulate. Hence the hybrid trajectory is forward complete across any infinite sequence of life-arcs.

Theological gloss. This remark is not used in the proof. The gate

$$\mathcal{G}^-_{\mathrm{recepta}} \geq R_c = C_\sigma - \sigma_0 - \lambda_0 \tag{149}$$

is the threshold past which Sophia can no longer be extinguished by resistance; in that regime the balance forces $\lambda^- \geq 0$, and strictly $\lambda^- > 0$ whenever $\sigma^- < C_\sigma$. Its complement, $\lambda^- \to 0$, is the self-closing limit $I_{\mathrm{gate}} \to 0$: final impenitence. It resembles the oil of the wise virgins and the *donum perseverantiae*. The upper boundary $R_{\max}$ represents the opposite failure: an over-diluting or "cheap grace" regime.

17 Non-Zeno Estimate

Hybrid margin:

$$\bar{h}_* = \min\left\{m_{pb}, h_{3,*}, r_*, 1 - r^*, q^*, C_\sigma, \sqrt{D^*}\right\} > 0. \tag{150}$$

Bounded vector field:

$$\bar{M}_* < \infty. \tag{151}$$

Dwell-time lower bound:

$$\Delta t_j \geq \Delta t_* := \frac{\bar{h}_*}{\bar{M}_*} > 0. \tag{152}$$

Non-Zeno conclusion:

$$\sum_j \Delta t_j = +\infty. \tag{153}$$

18 Memory and Gate Bounds

Mensura kernel:

$$R(z) = \frac{z}{z+K}, \qquad z \geq 0, \qquad 0 \leq R(z) \leq 1. \tag{154}$$

Saturated gate memory:

$$\frac{H_K}{H_s + H_K} \in [0, 1) \qquad \forall H_K \geq 0. \tag{155}$$

Gate saturation:

$$g_{\mathrm{eff}} = e^{-\tau t} + (1 - e^{-\tau t})\frac{H_K}{H_s + H_K} \in [0, 1]. \tag{156}$$

Memory velocity bounds:

$$|\dot{H}_K| \le \eta\,[R((\dot{x}_0)_+) + R((\dot{x}_0)_-)] \le 2\eta. \tag{157}$$

Wound-memory velocity bound:

$$|\dot{W}| \le 2\eta. \tag{158}$$

Bounded-gain loop:

$$H_K \xrightarrow{\le 1} g_{\text{eff}} \xrightarrow{\le S_q^{\max}} S_q \to \lambda \to V, F \to u \to \dot{x}_0 \xrightarrow{R \le 1} \dot{H}_K. \tag{159}$$

Extended field bound:

$$\bar{M}_* < \infty \quad \text{including} \quad |\dot{H}_K|, |\dot{W}| \le 2\eta. \tag{160}$$

19 Asymptotic Cleansing and Alignment

Will–action alignment:

$$D_\Delta(t) \le D_\Delta(0)e^{-2a\mu_* t}, \qquad |h_1(t) - h_2(t)| \le |h_1(0) - h_2(0)|e^{-a\mu_* t} \to 0. \tag{161}$$

Resistance cleansing under persistent active margin:

$$u(t) \ge \Lambda_c + \eta \quad \forall t \ge 0 \quad \Longrightarrow \quad \Psi_\sigma(t) \le \Psi_\sigma(0)e^{-c_\sigma t} \quad \Longrightarrow \quad \sigma(t) \to 0. \tag{162}$$

Corollary (unconditional cleansing inside the active domain). Since $\mathcal{D}_A$ is positively invariant, one has $r(t) \ge r_*$ and $h_3(t) \ge h_{3,*}$ for all $t \ge 0$. Thus

$$r(t) + h_3(t) - 1 \ge r_* + h_{3,*} - 1 \ge \frac{\eta(1 - h_{3,*})}{\Lambda_c} > 0. \tag{163}$$

Equivalently,

$$u(t) \ge \Lambda_c + \eta \qquad \forall t \ge 0. \tag{164}$$

Therefore the hypothesis of the persistent-margin cleansing statement holds automatically for every trajectory with $x(0) \in \mathcal{D}_A$. Hence

$$\sigma(t) \to 0, \qquad h_1(t) - h_2(t) \to 0. \tag{165}$$

Open-balance asymptotic Sophia:

$$\lambda(t) = \sigma_0 + \lambda_0 + \mathcal{G}_{\text{recepta}}(t) - \sigma(t). \tag{166}$$

If $\mathcal{G}_{\text{recepta}}(\infty) < \infty$ and $\sigma(t) \to 0$:

$$\lambda(t) \to \sigma_0 + \lambda_0 + \mathcal{G}_{\text{recepta}}(\infty). \tag{167}$$

If $\liminf_{t\to\infty} \lambda(t) > 0$:

$$\Omega_P(t) \to +\infty. \tag{168}$$

Epektatic dilution of S_q:

$$\int_0^\infty \lambda(t)\,dt = +\infty \quad \Longrightarrow \quad \Omega_P(t) \to +\infty \quad \Longrightarrow \quad S_q(t) \to 0. \tag{169}$$

Proof of the last implication:

$$0 \le S_q(t) \le \frac{\zeta_0}{\Omega_P(t)} \to 0. \tag{170}$$

20 Numerical Witness Formulae

One admissible parameter point recorded in the Lyapunov file:

$$
\begin{aligned}
&a = 3.494, \quad c = 0.429, \quad \alpha = 0.529, \quad \delta = 0.898, \quad \rho = 0.545, \\
&\zeta_0 = 0.560, \quad h_{3,*} = 0.622, \quad r_* = r_- = 0.531, \quad r^* = r_+ = 0.892, \\
&\varepsilon = 0.006, \quad d = 0.193, \qquad \Lambda_c = 1 \quad \text{(normalization).}
\end{aligned} \tag{171}
$$

Derived quantities:

$$
\begin{aligned}
e_{\max} &= \varepsilon(1 - h_{3,*})^2 = 0.0009, \\
S_q^{\max} &= \zeta_0(1 - h_{3,*}) = 0.212, \\
q^*_{\text{new}} &= \sqrt{\frac{\delta r_+ + S_q^{\max}}{\alpha}} = 1.383, \\
q^* &= 1.05 q^*_{\text{new}} = 1.452.
\end{aligned} \tag{172}
$$

Active margin:

$$
r_- + h_{3,*} - 1 = 0.153 > 0, \qquad \frac{u}{\Lambda_c} \approx 1.40. \tag{173}
$$

Thus the witness lies in the Transformation region.

All five parametric walls hold strictly; the worst-case slack is

$$
0.084. \tag{174}
$$

21 Domain Stability Theorem (Complete Proof)

This final section consolidates the preceding formulae into a single rigorous statement: the active domain is positively invariant, forward complete, never contacts the death-boundary, and exhibits asymptotic cleansing and alignment, with admissible resets continuing the trajectory without Zeno accumulation. The dissipative step is Proposition 1; the light-filter constant is Lemma 1.

21.1 Setup recap and hypotheses

We use the normalized flow (Section 20 notation; $\Lambda_c = 1$),

$$
\begin{aligned}
\dot h_1 &= \mu(a h_2 + c q) - (\rho + \alpha q) h_1, & \dot h_2 &= \mu(a h_1 + c q) - (\rho + \alpha q) h_2, \\
\dot q &= \delta m T - \alpha q^2 + S_q, & \dot h_3 &= \alpha q (1 - h_3), \\
\dot \sigma &= -\delta\, \Omega_P\, m T, & S_q &\in [0, S_q^{\max}],
\end{aligned} \tag{175}
$$

with $\Omega_P = 1/(1 - h_3)$, $T = \tanh(\kappa\sigma)$, $\mu = 1 - r$, $e = \varepsilon(1 - h_3)^2$, $r^2 = h_1 h_2 + e$, $m = r + h_3 - 1$, and the Lyapunov functional $\mathcal{L}_{\text{VFS}} = A_\sigma \Psi_\sigma + A_\Delta D_\Delta + A_q q^2 + A_r B(r)$ with $\Psi_\sigma = \frac{1}{\kappa} \ln\cosh \kappa\sigma$, $D_\Delta = \frac{1}{2}(h_1 - h_2)^2$, $B(r) = -\ln(1 - r)$, weights $A_\bullet > 0$.

Definition 1 (Active domain)**.**

$$
\mathcal{D}_A = \left\{ (\sigma, h_1, h_2, q, h_3) : \begin{array}{l} \sigma \in [0, C_\sigma], \quad q \in [0, q^*], \quad h_3 \in [h_{3,*}, 1), \quad r \in [r_*, r^*], \\ D_\Delta \le D^*, \quad h_1, h_2 \ge 0 \end{array} \right\}.
$$

Here $\mu_* = 1 - r^*$, $\Omega_{min} = 1/(1 - h_{3,*})$, $m_* = r_* + h_{3,*} - 1$, $e_{\max} = \varepsilon(1 - h_{3,*})^2$, $d_* = \sqrt{2D^*}$, and $q^* = 1.05\sqrt{(\delta r^* + S_q^{\max})/\alpha}$.

Standing Hypothesis 1 (Parameters)**.** $a, c, \alpha, \delta, \gamma, \zeta_0, \kappa, \varepsilon > 0$, $\rho \geq 0$.

Standing Hypothesis 2 (Walls)**.** $r_*^2 > e_{\max}$; $m_* > 0$; $\alpha(q^*)^2 \geq \delta r^* + S_q^{\max}$;

$$(1-r^*)\left[a(d_*^2+2r^{*2})+cq^*\sqrt{d_*^2+4r^{*2}}\right] \leq 2\rho(r^{*2}-e_{\max}), \quad a(1-r_*)(r_*^2-e_{\max}) \geq (\rho+\alpha q^*)r_*^2. \tag{176}$$

Standing Hypothesis 3 (Reset window)**.** At each death-event $\max\{R_c, R_{\min}^{\text{ceil}}\} \leq \mathcal{G}_{\text{recepta}}^- \leq R_{\max}$, $R_{\max} = \Omega_P^-(r^- - r_*)/(\kappa_R r_*)$, for the jump $\sigma^+ = q_R\sigma^-$, $\lambda^+ = \lambda^- + (1-q_R)\sigma^-$, $\Omega_P^+ = \Omega_P^- + \kappa_R\mathcal{G}_{\text{recepta}}^-$, $(V,F)^+ = (V,F)^-$, $q_R \in (0,1)$.

21.2 Identities, regularity, invariance, completeness

Lemma 2 (Identities)**.** *Along* (175)*,* $2r\dot{r} = \mathcal{R}_{\text{rad}} := \mu[a(h_1^2+h_2^2)+cq(h_1+h_2)]-2\rho(r^2-e)-2\alpha qr^2$ *and* $\dot{D}_\Delta = -2(a\mu+\rho+\alpha q)D_\Delta$*.*

Proof. Differentiate $r^2 = h_1h_2 + e(h_3)$ and use (175); the h_1h_2 and e contributions combine via $h_1h_2 = r^2 - e$ and $e'(h_3)\dot{h}_3 = -2\alpha q\, e$. For D_Δ, $\frac{d}{dt}(h_1-h_2) = -(a\mu+\rho+\alpha q)(h_1-h_2)$. □

Lemma 3 (Regularity)**.** *The field* (175) *is* C^1 *near* $\mathcal{D}_A$ *(since* $VF+\varepsilon \geq \varepsilon > 0$ *and* $h_3 < 1$*), giving unique maximal solutions.*

Lemma 4 (Positive invariance — whole-face Nagumo)**.** *Under Hypotheses 1–2,* $\mathcal{D}_A$ *is positively invariant.*

Proof. On $\mathcal{D}_A$: $m \geq m_* > 0$, $\mu \geq \mu_*$, $\Omega_P \geq \Omega_{min}$, $T \in [0,1]$. We verify sub-tangentiality face by face (Nagumo). *σ=C_σ:* $\dot{\sigma} = -\delta\Omega_P m\tanh(\kappa C_\sigma) < 0$. *$\sigma$=0:* $\dot{\sigma} = 0$ (tangent). *h_3=$h_{3,*}$:* $\dot{h}_3 = \alpha q(1-h_3) \geq 0$ (h_3 non-decreasing). *D_Δ=D^*:* $\dot{D}_\Delta \leq 0$ by Lemma 2. *q=0:* $\dot{q} = \delta mT + S_q \geq 0$ (no lower q-floor needed). *q=q^*:* $\dot{q} \leq \delta r^* + S_q^{\max} - \alpha(q^*)^2 \leq 0$ by the q-wall. *r=r^*:* writing $\mathcal{R}_{\text{rad}} = P - N$ with $P = \mu_*[a(h_1^2+h_2^2)+cq(h_1+h_2)]$, $N = 2\rho(r^{*2}-e)+2\alpha qr^{*2}$, the face bounds $(h_1-h_2)^2 \leq d_*^2$, $0 \leq e \leq e_{\max}$, $0 \leq q \leq q^*$ give $\sup P \leq \mu_*[a(d_*^2+2r^{*2})+cq^*\sqrt{d_*^2+4r^{*2}}]$ and $\inf N \geq 2\rho(r^{*2}-e_{\max})$, so $\sup \mathcal{R}_{\text{rad}} \leq \sup P - \inf N \leq 0$ by the upper radial wall; hence $\dot{r} \leq 0$. *r=r_*:* symmetrically $\inf P \geq 2a(1-r_*)(r_*^2-e_{\max})$, $\sup N \leq 2(\rho+\alpha q^*)r_*^2$, so $\inf \mathcal{R}_{\text{rad}} \geq 0$ by the lower radial wall; hence $\dot{r} \geq 0$. All faces sub-tangential $\Rightarrow$ $\mathcal{D}_A$ invariant. □

Lemma 5 (Forward completeness)**.** *Every solution from* $\mathcal{D}_A$ *exists on* $[0,\infty)$*, with* $1-h_3(t) \geq (1-h_3(0))e^{-\alpha q^* t} > 0$*, so* $h_3 < 1$ *and* $\Omega_P < \infty$ *on finite intervals.*

Proof. By Lemma 4 the orbit stays in $\mathcal{D}_A$ (all coordinates but Ω_P bounded); from $\dot{h}_3 = \alpha q(1-h_3)$, $q \leq q^*$, integrate $\frac{d}{dt}(1-h_3) \geq -\alpha q^*(1-h_3)$. No finite-time blow-up, so the orbit extends to $[0,\infty)$. □

21.3 Dissipation, alignment, cleansing, barrier

The dissipative estimate $\dot{\mathcal{L}_{\text{VFS}}} \leq C - C_1\mathcal{L}_{\text{VFS}}$ with $C_1 = \min\{\delta\Omega_{min}m_*c_\Psi, 2a\mu_*\} > 0$ is Proposition 1 (using Lemma 1).

Lemma 6 (Alignment)**.** $D_\Delta(t) \leq D_\Delta(0)e^{-2a\mu_* t} \to 0$.

Proof. $\dot{D}_\Delta = -2(a\mu+\rho+\alpha q)D_\Delta \leq -2a\mu_*D_\Delta$; integrate. □

Lemma 7 (Cleansing). *$\sigma(t) \to 0$ as $t \to \infty$.*

Proof. On $\mathcal{D}_A$, $\dot{\sigma} = -\delta\Omega_P m \tanh(\kappa\sigma) \le -\beta \tanh(\kappa\sigma)$, $\beta = \delta\Omega_{min} m_* > 0$. Thus σ is nonincreasing, bounded below by 0, hence $\sigma \to \sigma_\infty \ge 0$; comparison with $\dot{y} = -\beta \tanh(\kappa y)$ (whose only, globally attracting, equilibrium is 0) forces $\sigma_\infty = 0$. □

Corollary 1 (Death-boundary never contacted). *For all $t \ge 0$,*

$$r(t) \le r^* < 1, \qquad \mathcal{L}_{\mathrm{VFS}}(t) \le \max\{\mathcal{L}_{\mathrm{VFS}}(0), C/C_1\}.$$

Consequently $u = r\Omega_P < \Omega_P$ always.

21.4 Hybrid layer

Lemma 8 (Reset admissibility). *Under Hypothesis 3, $x^- \in \mathcal{D}_A \Rightarrow \mathfrak{R}(x^-) \in \mathcal{D}_A$, with $r^+ \le r^-$, $\sigma^+ \le \sigma^-$, $D_\Delta^+ \le D_\Delta^-$, $h_3^+ \ge h_3^-$.*

Proof. (V, F) preserved $\Rightarrow u^+ = u^-$, $\Omega_P^+ = \Omega_P^- + \kappa_R \mathcal{G}^-_{\text{recepta}} \ge \Omega_P^-$, so $r^+ = r^- \Omega_P^- / \Omega_P^+ \le r^-$, $h_3^+ \ge h_3^-$, $\sigma^+ = q_R \sigma^- \le \sigma^-$, $D_\Delta^+ = (\Omega_P^-/\Omega_P^+)^2 D_\Delta^- \le D_\Delta^-$. The window upper bound $\mathcal{G}^-_{\text{recepta}} \le R_{\max}$ is equivalent to $r^+ \ge r_*$. Since $x^- \in \mathcal{D}_A$ gives $\lambda^- \ge 0$, one has $q^+ \ge 0$; and the ceiling-floor condition gives $q^+ \le q^*$. Thus $q^+ \in [0, q^*]$. All constraints of $\mathcal{D}_A$ hold; the margin $m^+ \ge m_*$ follows from $r^+ \ge r_*$, $h_3^+ \ge h_{3,*}$. □

Lemma 9 (Non-Zeno). *The field is bounded on $\mathcal{D}_A$ ($\sup \|f\| = \bar{M}_* < \infty$) and resets land with a uniform interior margin $\bar{h}_* > 0$; hence consecutive resets are separated by $\Delta t_j \ge \bar{h}_*/\bar{M}_* > 0$, so the hybrid trajectory is non-Zeno and forward complete.*

21.5 Main theorem

Theorem 1 (Domain stability of $\mathcal{D}_A$). *Assume Hypotheses 1–3. Then for every $x(0) \in \mathcal{D}_A$:* (i) *$x(t) \in \mathcal{D}_A$ $\forall t \ge 0$;* (ii) *the solution exists on $[0, \infty)$ with $h_3(t) < 1$, $\Omega_P(t) < \infty$ on finite intervals;* (iii) *$\mathcal{L}_{\mathrm{VFS}}(t) \le \max\{\mathcal{L}_{\mathrm{VFS}}(0), C/C_1\}$, $C_1 > 0$, and $r(t) \le r^* < 1$ (death-boundary never contacted);* (iv) *$\sigma(t) \to 0$ and $D_\Delta(t) \to 0$;* (v) *under admissible resets the hybrid trajectory stays in $\mathcal{D}_A$, is non-Zeno, and forward complete. Schematically,*

$$\boxed{\textit{Hyp. } 1\text{–}2 \Rightarrow \mathcal{D}_A \textit{ invariant} \Rightarrow \textit{complete} \Rightarrow \dot{\mathcal{L}}_{\mathrm{VFS}} \le C - C_1 \mathcal{L}_{\mathrm{VFS}} \Rightarrow r < 1 \Rightarrow \sigma, D_\Delta \to 0.} \quad (177)$$

Proof. (i) Lemma 4; (ii) Lemma 5; (iii) Proposition 1 with Corollary 1; (iv) Lemmas 6, 7, using the invariant lower bounds $\mu \ge \mu_*$, $m \ge m_*$ from (i); (v) Lemmas 8, 9. □

Remark (Nature of the stability). This is *set* stability with *partial* convergence: $\mathcal{D}_A$ is invariant and $(\sigma, D_\Delta) \to 0$, while (q, Ω_P) need not converge — $\dot{\Omega}_P = \alpha\lambda > 0$ for $\lambda > 0$, so $\Omega_P \to \infty$ under sustained inflow (*epektasis*). There is no equilibrium; $\mathcal{L}_{\mathrm{VFS}}$ is a boundedness-and-barrier certificate, not a function vanishing at a rest point. The conclusion is strictly domain-conditional ($x(0) \in \mathcal{D}_A$), not global or automatic.

Remark (Folding-compatible extension). If the regular folding conditions of the preceding sections are imposed in addition to Hypotheses 1–3, the same domain stability statement holds for the extended functional

$$\mathcal{L}_{\text{ext}} = \mathcal{L}_{\mathrm{VFS}} + w_{\text{fold}} \mathcal{V}_{\text{fold}}. \quad (178)$$

The original theorem is recovered by setting $q_{\text{fold}} = 0$ and $\lambda_{\text{form}} = 0$. In the refined model, folding adds stabilization by form-transformation: resistance still cleanses, will and action still align, and Sophianic excess may additionally relax into Vessel-form without breaking the Open-Gate Lyapunov domain.

Remark (Closed limit). At $\zeta_0 = 0$: $S_q = I_{\text{gate}} = 0$, $\dot{\lambda} = -\dot{\sigma}$, and $\sigma + \lambda = \sigma_0 + \lambda_0$ (exact conservation). The Open-Gate model adds only the bounded S_q in $\dot{q}$: it strengthens the walls (a larger q^*) but leaves the dissipative core unchanged.

Remark (Grace enters C, never C_1). I_{gate} enters (175) only through S_q, feeding the bounded blocks of Proposition 1; it never enters κ_σ or κ_Δ. Grace raises C by at most $\Delta C_{\text{gate}} \le 2A_q q^* S_q^{\max}$ and can neither manufacture nor destroy $C_1 > 0$: the transforming structure is sustained, not replaced. Moreover $\inf C_1 = 0$ at the admissibility boundary ($m_* \to 0^+$ or $\mu_* \to 0^+$): a state pinned on the active margin has zero cleansing rate — stability holds, transfiguration stalls.